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THERMOELECTRIC MODULE AND REFRIGERATOR COMPRISING SAME

Abstract

A thermoelectric module according to an embodiment of the present invention may comprise: a cold sink; a thermoelectric element having a heat absorption surface coupled to the cold sink; a heat sink coupled to a heating surface of the thermoelectric element to dissipate heat transferred from the cold sink to the outside of the thermoelectric element; and a sealing cover for connecting the edge of the cold sink and the edge of the heat sink to surround the thermoelectric element, wherein the cold sink, the heat sink, and the thermoelectric element may be integrally formed by the sealing cover.

In addition, the thermoelectric element may be a cascade type thermoelectric element in which two thermoelectric elements having the same or different specifications are coupled to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of U.S. application Ser. No. 17/434,708 filed on Apr. 5, 2022, now allowed, which is a National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2020/004071, with an international filing date of Mar. 25, 2020, which claims the benefit of Korean Patent Application No. 10-2019-0076584, filed on Jun. 26, 2019 all of which are incorporated by reference in their entirety herein.

TECHNICAL FIELD

[0002] The present invention relates to a thermoelectric module and a refrigerator including the same.

BACKGROUND ART

[0003] In general, a refrigerator is a home appliance for storing food at a low temperature, and includes a refrigerating compartment for storing food in a refrigerated state in a range of 3° C. and a freezing compartment for storing food in a frozen state in a range of −20° C.

[0004] However, when food such as meat or seafood is stored in the frozen state in the existing freezing compartment, moisture in cells of the meat or seafood are escaped out of the cells in the process of freezing the food at the temperature of −20° C., and thus, the cells are destroyed, and taste of the food is changed during an unfreezing process.

[0005] However, if a temperature condition of the storage compartment is set to a cryogenic state that is significantly lower than the current temperature of the freezing temperature. Thus, when the food quickly passes through a freezing point temperature range while the food is changed in the frozen state, the destruction of the cells may be minimized, and as a result, even after the unfreezing, the meat quality and the taste of the food may return to close to the state before the freezing. The cryogenic temperature may be understood to mean a temperature in a range of −45° C. to −50° C.

[0006] For this reason, in recent years, the demand for a refrigerator equipped with a deep freezing compartment that is maintained at a temperature lower than a temperature of the freezing compartment is increasing.

[0007] In order to satisfy the demand for the deep freezing compartment, there is a limit to the cooling using an existing refrigerant. Thus, an attempt is made to lower the temperature of the deep freezing compartment to a cryogenic temperature by using a thermoelectric module (TEM).

[0008] Korean Patent Publication No. 2018-0105572 (Sep. 28, 2018) that is a prior art discloses a refrigerator having the form of a bedside table, in which a storage compartment has a temperature lower than the room temperature by using a thermoelectric module.

[0009] However, in the case of the refrigerator using the thermoelectric module disclosed in the prior art, since a heat generation surface of the thermoelectric module is configured to be cooled by heat-exchanged with indoor air, there is a limitation in lowering a temperature of the heat absorption surface.

[0010] In detail, in the thermoelectric module, when supply current increases, a temperature

difference between the heat absorption surface and the heat generation surface tends to increase to a certain level. However, due to characteristics of the thermoelectric element made of a semiconductor element, when the supply current increases, the semiconductor acts as resistance to increase in self-heat amount. Then, there is a problem that heat absorbed from the heat absorption surface is not transferred to the heat generation surface quickly.

[0011] In addition, if the heat generation surface of the thermoelectric element is not sufficiently cooled, a phenomenon in which the heat transferred to the heat generation surface flows back toward the heat absorption surface occurs, and a temperature of the heat absorption surface also rises.

[0012] In the case of the thermoelectric module disclosed in the prior art, since the heat generation surface is cooled by the indoor air, there is a limit that the temperature of the heat generation surface is not lower than a room temperature.

[0013] In a state in which the temperature of the heat generation surface is substantially fixed, the supply current has to increase to lower the temperature of the heat absorption surface, and then efficiency of the thermoelectric module is deteriorated.

[0014] In addition, if the supply current increases, a temperature difference between the heat absorption surface and the heat generation surface increases, resulting in a decrease in the cooling capacity of the thermoelectric module.

DISCLOSURE OF THE INVENTION

Technical Problem

[0015] The present invention has been proposed to improve the above-described limitations.

Technical Solution

[0016] A thermoelectric module according to an embodiment of the present invention for achieving the above object include: a cold sink; a thermoelectric element having a heat absorption surface coupled to the cold sink; a heat sink coupled to a heat generation surface of the thermoelectric element to dissipate heat transferred from the cold sink to the outside of the thermoelectric element; and a sealing cover for connecting an edge of the cold sink and the edge of the heat sink to surround the thermoelectric element, wherein the cold sink, the heat sink, and the thermoelectric element are integrally formed by the sealing cover.

[0017] In addition, the thermoelectric element may be a cascade type thermoelectric element in which two thermoelectric elements having the same or different specifications are coupled to each other.

[0018] In addition, in the cascade type thermoelectric element, a heat dissipation sheet may be applied instead of the conventional ceramic substrate, and a metal block may be interposed at a bonding portion of two thermoelectric elements, so that a thickness of the thermoelectric module is significantly reduced.

[0019] In addition, the thermoelectric module or the cascade type thermoelectric module in which the heat sink and the cold sink are integrally disposed may be used to cool the deep freezing compartment to an ultra-low temperature of about -50°C .

Advantageous Effects

[0020] The thermoelectric module having the above-described constitutions and the refrigerator including the same according to the embodiment may have following effects.

[0021] First, since the ceramic substrate provided on both the ends of the thermoelectric element is removed, there may be the advantage in that the thickness of the thermoelectric module is reduced.

[0022] Second, since the material of the plating layer of the electrode portion of the semiconductor element constituting the thermoelectric element is changed from tin (Sn) to gold (Au), the solder wetting may be improved to improve the soldering stability of the semiconductor element.

[0023] Third, as the ceramic substrate is removed, there may be the advantage in that the coating process of the thermal grease for bonding the ceramic substrate to the heat sink or the cold sink may be removed.

[0024] Fourth, instead of the relatively thick ceramic substrate, the heat dissipation sheet may be interposed in both the ends of the semiconductor device, and the heat dissipation sheet and the heat sink, and the heat dissipation sheet and the cold sink may be integrally coupled to each other to significantly improve the heat transfer effect.

[0025] Fifth, the process of sealing the side surface of the thermoelectric element may be eliminated, and the sealing cover directly connecting the heat sink to the cold sink may be provided to simplify the manufacturing process and reduce the manufacturing cost.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a view illustrating a refrigerant circulation system of a refrigerator according to an embodiment of the present invention.

[0027] FIG. 2 is a perspective view illustrating structures of a freezing compartment and a deep freezing compartment of the refrigerator according to an embodiment of the present invention.

[0028] FIG. 3 is a longitudinal cross-sectional view taken along line 3-3 of FIG. 2.

[0029] FIG. 4 is a graph illustrating a relationship of cooling capacity with respect to an input voltage and a Fourier effect.

[0030] FIG. 5 is a graph illustrating a relationship of efficiency with respect to an input voltage and a Fourier effect.

[0031] FIG. 6 is a graph illustrating a relationship of cooling capacity and efficiency according to a voltage.

[0032] FIG. 7 is a view illustrating a reference temperature line for controlling a refrigerator according to a change in load inside the refrigerator.

[0033] FIG. 8 is a cross-sectional view of a thermoelectric module according to a first embodiment of the present invention.

[0034] FIG. 9 is a cross-sectional view of a thermoelectric module according to a second embodiment of the present invention.

[0035] FIG. 10 is a cross-sectional view of a thermoelectric module according to a third embodiment of the present invention.

[0036] FIG. 11 is a cross-sectional view of a thermoelectric module according to a fourth embodiment of the present invention.

[0037] FIG. 12 is a cross-sectional view of a thermoelectric module according to a fifth embodiment of the present invention.

[0038] FIG. 13 is a cross-sectional view of a thermoelectric module according to a sixth embodiment of the present invention.

MODE FOR CARRYING OUT THE INVENTION

[0039] Hereinafter, a thermoelectric module and a refrigerator including the same according to embodiments will be described in detail with reference to the accompanying drawings.

[0040] In the present invention, a storage compartment that is cooled by a first cooling device and controlled to a predetermined temperature may be defined as a first storage compartment.

[0041] In addition, a storage compartment that is cooled by a second cooling device and is controlled to a temperature lower than that of the first storage compartment may be defined as a second storage compartment.

[0042] In addition, a storage compartment that is cooled by the third cooling device and is controlled to a temperature lower than that of the second storage compartment may be defined as a third storage compartment.

[0043] The first cooling device for cooling the first storage compartment may include at least one of a first evaporator or a first thermoelectric module including a thermoelectric element. The first

evaporator may include a refrigerating compartment evaporator to be described later.

[0044] The second cooling device for cooling the second storage compartment may include at least one of a second evaporator or a second thermoelectric module including a thermoelectric element. The second evaporator may include a freezing compartment evaporator to be described later.

[0045] The third cooling device for cooling the third storage compartment may include at least one of a third evaporator or a third thermoelectric module including a thermoelectric element.

[0046] In the embodiments in which the thermoelectric module is used as a cooling means in the present specification, it may be applied by replacing the thermoelectric module with an evaporator, for example, as follows. [0047] (1) “Cold sink of thermoelectric module”, “heat absorption surface of thermoelectric module” or “heat absorption side of thermoelectric module” may be interpreted as “evaporator or one side of the evaporator”. [0048] (2) “Heat absorption side of thermoelectric module” may be interpreted as the same meaning as “cold sink of thermoelectric module” or “heat absorption side of thermoelectric module”. [0049] (3) A controller “applies or cuts off a constant voltage to the thermoelectric module” may be interpreted as any one of being controlled to “supply or block a refrigerant to the evaporator”, “control a switching valve to be opened or closed”, or “control a compressor to be turned on or off”. [0050] (4) “Controlling the constant voltage applied to the thermoelectric module to increase or decrease” by the controller may be interpreted as any one of “controlling an amount or flow rate of the refrigerant flowing in the evaporator to increase or decrease”, “controlling allowing an opening degree of the switching valve to increase or decrease”, or “controlling an output of the compressor to increase or decrease”. [0051] (5) “Controlling a reverse voltage applied to the thermoelectric module to increase or decrease” by the controller is interpreted as “controlling a voltage applied to the defrost heater adjacent to the evaporator to increase or decrease”.

[0052] In the present specification, “storage compartment cooled by the thermoelectric module” is defined as a storage compartment A, and “fan located adjacent to the thermoelectric module so that air inside the storage compartment A is heat-exchanged with the heat absorption surface of the thermoelectric module” may be defined as “storage compartment fan A”.

[0053] Also, a storage compartment cooled by the cooling device while constituting the refrigerator together with the storage compartment A may be defined as “storage compartment B”.

[0054] In addition, a “cooling device compartment” may be defined as a space in which the cooling device is disposed, in a structure in which the fan for blowing cool air generated by the cooling device is added, the cooling device compartment may be defined as including a space in which the fan is accommodated, and in a structure in which a passage for guiding the cold air blown by the fan to the storage compartment or a passage through which defrost water is discharged is added may be defined as including the passages.

[0055] In the present invention, as an example, the first storage compartment may include a refrigerating compartment that is capable of being controlled to an above zero temperature by the first cooling device.

[0056] In addition, the second storage compartment may include a freezing compartment that is capable of being controlled to a temperature below zero by the second cooling device.

[0057] In addition, the third storage compartment may include a deep freezing compartment that is capable of being maintained at a cryogenic temperature or an ultrafreezing temperature by the third cooling device.

[0058] In the present invention, a case in which all of the third to third storage compartments are controlled to a temperature below zero, a case in which all of the first to third storage compartments are controlled to a above zero temperature, and a case in which the first and second storage compartments are controlled to the above zero temperature, and the third storage compartment is controlled to the temperature below zero are not excluded.

[0059] Hereinafter, as an example, the description is limited to the case in which the first storage compartment is the refrigerating compartment, the second storage compartment is the freezing

compartment, and the third storage compartment is the deep freezing compartment.

[0060] FIG. 1 is a view illustrating a refrigerant circulation system of a refrigerator according to an embodiment of the present invention.

[0061] Referring to FIG. 1, a refrigerant circulation system **10** according to an embodiment of the present invention includes a compressor **11** that compresses a refrigerant into a high-temperature and high-pressure gaseous refrigerant, a condenser **12** that condenses the refrigerant discharged from the compressor **11** into a high-temperature and high-pressure liquid refrigerant, an expansion valve that expands the refrigerant discharged from the condenser **12** into a low-temperature and low-pressure two-phase refrigerant, and an evaporator that evaporates the refrigerant passing through the expansion valve into a low-temperature and low-pressure gaseous refrigerant. The refrigerant discharged from the evaporator flows into the compressor **11**. The above components are connected to each other by a refrigerant pipe to constitute a closed circuit.

[0062] In detail, the expansion valve may include a refrigerator compartment expansion valve **14** and a freezing compartment expansion valve **15**. The refrigerant pipe is divided into two branches at an outlet side of the condenser **12**, and the refrigerating compartment expansion valve **14** and the freezing compartment expansion valve **15** are respectively connected to the refrigerant pipe that is divided into the two branches. That is, the refrigerating compartment expansion valve **14** and the freezing compartment expansion valve **15** are connected in parallel at the outlet of the condenser **12**.

[0063] A switching valve **13** is mounted at a point at which the refrigerant pipe is divided into the two branches at the outlet side of the condenser **12**. The refrigerant passing through the condenser **12** may flow through only one of the refrigerating compartment expansion valve **14** and the freezing compartment expansion valve **15** by an operation of adjusting an opening degree of the switching valve **13** or may flow to be divided into both sides.

[0064] The switching valve **13** may be a three-way valve, and a flow direction of the refrigerant is determined according to an operation mode. Here, one switching valve such as the three-way valve may be mounted at an outlet of the condenser **12** to control the flow direction of the refrigerant, or alternatively, the switching valves are mounted at inlet sides of a refrigerator compartment expansion valve **14** and a freezing compartment expansion valve **15**, respectively.

[0065] As a first example of an evaporator arrangement manner, the evaporator may include a refrigerating compartment evaporator **16** connected to an outlet side of the refrigerating compartment expansion valve **14** and a heat sink **24** and a freezing compartment evaporator **17**, which are connected in series to an outlet side of the freezing compartment expansion valve **15**. The heat sink **24** and the freezing compartment evaporator **17** are connected in series, and the refrigerant passing through the freezing compartment expansion valve passes through the heat sink **24** and then flows into the freezing compartment evaporator **17**.

[0066] Second, as a second example, the heat sink **24** may be disposed at an outlet side of the freezing compartment evaporator **17** so that the refrigerant passing through the freezing compartment evaporator **17** flows into the heat sink **24**.

[0067] Third, as a third example, a structure in which the deep freezing compartment evaporator **24** and the freezing compartment evaporator **17** are connected in parallel at an outlet end of the freezing compartment expansion valve **15** is not excluded.

[0068] Although the heat sink **24** is the evaporator, it is provided for the purpose of cooling a heat generation surface of the thermoelectric module to be described later, not for the purpose of heat-exchange with the cold air of the deep freezing compartment.

[0069] In each of the three examples described above with respect to the arrangement manner of the evaporator, a complex system of a first refrigerant circulation system, in which the switching valve **13**, the refrigerating compartment expansion valve **14**, and the refrigerating compartment evaporator **16** are removed, and a second refrigerant circulation system constituted by the refrigerating compartment cooling evaporator, the refrigerating compartment cooling expansion

valve, the refrigerating compartment cooling condenser, and a refrigerating compartment cooling compressor is also possible.

[0070] Here, the condenser constituting the first refrigerant circulation system and the condenser constituting the second refrigerant circulation system may be independently provided, and a complex condenser which is provided as a single body and in which the refrigerant is not mixed may be provided.

[0071] The refrigerant circulation system of the refrigerator having the two storage compartments including the deep freezing compartment may be configured only with the first refrigerant circulation system.

[0072] Hereinafter, as an example, the description will be limited to a structure in which the heat sink and the freezing compartment evaporator **17** are connected in series.

[0073] A condensing fan **121** is mounted adjacent to the condenser **12**, a refrigerating compartment fan **161** is mounted adjacent to the refrigerating compartment evaporator **16**, and a freezing compartment fan **171** is mounted adjacent to the freezing compartment evaporator **17**.

[0074] A refrigerating compartment maintained at a refrigerating temperature by cold air generated by the refrigerating compartment evaporator **16**, a freezing compartment maintained at a freezing temperature by cold air generated by the freezing compartment evaporator **16**, and a deep freezing compartment **202** maintained at a cryogenic or ultrafreezing temperature by a thermoelectric module to be described later are formed inside the refrigerator provided with the refrigerant circulation system according to the embodiment of the present invention. The refrigerating compartment and the freezing compartment may be disposed adjacent to each other in a vertical direction or horizontal direction and are partitioned from each other by a partition wall. The deep freezing compartment may be provided at one side of the inside of the freezing compartment, but the present invention includes the deep freezing compartment provided at one side of the outside of the freezing compartment. In order to block the heat exchange between the cold air of the deep freezing compartment and the cold air of the freezing compartment, the deep freezing compartment **202** may be partitioned from the freezing compartment by a deep freezing case **201** having the high thermal insulation performance.

[0075] In addition, the thermoelectric module includes a thermoelectric element **40** having one side through which heat is absorbed and the other side through which heat is released when power is supplied, a cold sink **22** mounted on the heat absorption surface of the thermoelectric element **40**, a heat sink **24** mounted on the heat generation surface of the thermoelectric element **40**, and an insulator **23** that blocks heat exchange between the cold sink **22** and the heat sink **24**.

[0076] The insulator **23** may not be required according to an example of the thermoelectric element **40** described below.

[0077] Here, the heat sink **24** is an evaporator that is in contact with the heat generation surface of the thermoelectric element **40**. That is, the heat transferred to the heat generation surface of the thermoelectric element **40** is heat-exchanged with the refrigerant flowing inside the heat sink **24**. The refrigerant flowing along the inside of the heat sink **24** and absorbing heat from the heat generation surface of the thermoelectric element **40** is introduced into the freezing compartment evaporator **17**.

[0078] In addition, a cooling fan may be provided in front of the cold sink **22**, and the cooling fan may be defined as the deep freezing compartment fan **25** because the fan is disposed behind the inside of the deep freezing compartment.

[0079] The cold sink **22** is disposed behind the inside of the deep freezing compartment **202** and configured to be exposed to the cold air of the deep freezing compartment **202**. Thus, when the deep freezing compartment fan **25** is driven to forcibly circulate cold air in the deep freezing compartment **202**, the cold sink **22** absorbs heat through heat-exchange with the cold air in the deep freezing compartment and then is transferred to the heat absorption surface of the thermoelectric element **40**. The heat transferred to the heat absorption surface is transferred to the heat generation

surface of the thermoelectric element **40**.

[0080] The heat sink **24** functions to absorb the heat absorbed from the heat absorption surface of the thermoelectric element **40** and transferred to the heat generation surface of the thermoelectric element **40** again to release the heat to the outside of the thermoelectric module **20**.

[0081] FIG. **2** is a perspective view illustrating structures of the freezing compartment and the deep freezing compartment of the refrigerator according to an embodiment of the present invention, and FIG. **3** is a longitudinal cross-sectional view taken along line **3-3** of FIG. **2**.

[0082] Referring to FIGS. **2** and **3**, the refrigerator according to an embodiment of the present invention includes an inner case **101** defining the freezing compartment **102** and a deep freezing unit **200** mounted at one side of the inside of the freezing compartment **102**.

[0083] In detail, the inside of the refrigerating compartment is maintained to a temperature of about 3°C ., and the inside of the freezing compartment **102** is maintained to a temperature of about -18°C ., whereas a temperature inside the deep freezing unit **200**, i.e., an internal temperature of the deep freezing compartment **202** has to be maintained to about -50°C . Therefore, in order to maintain the internal temperature of the deep freezing compartment **202** at a cryogenic temperature of -50°C ., an additional freezing means such as the thermoelectric module **20** is required in addition to the freezing compartment evaporator.

[0084] In more detail, the deep freezing unit **200** includes a deep freezing case **201** that forms a deep freezing compartment **202** therein, a deep freezing compartment drawer **203** slidably inserted into the deep freezing case **201**, and a thermoelectric module **20** mounted on a rear surface of the deep freezing case **201**.

[0085] Instead of applying the deep freezing compartment drawer **203**, a structure in which a deep freezing compartment door is connected to one side of the front side of the deep freezing case **201**, and the entire inside of the deep freezing compartment **201** is configured as a food storage space is also possible.

[0086] In addition, the rear surface of the inner case **101** is stepped backward to form a freezing evaporation compartment **104** in which the freezing compartment evaporator **17** is accommodated. In addition, an inner space of the inner case **101** is divided into the freezing evaporation compartment **104** and the freezing compartment **102** by the partition wall **103**. The thermoelectric module **20** is fixedly mounted on a front surface of the partition wall **103**, and a portion of the thermoelectric module **20** passes through the deep freezing case **201** and is accommodated in the deep freezing compartment **202**.

[0087] In detail, the heat sink **24** constituting the thermoelectric module **20** may be an evaporator connected to the freezing compartment expansion valve **15** as described above. A space in which the heat sink **24** is accommodated may be formed in the partition wall **103**.

[0088] Since the two-phase refrigerant cooled to a temperature of about -18°C . to -20°C . while passing through the freezing compartment expansion valve **15** flows inside the heat sink **24**, a surface temperature of the heat sink **24** may be maintained to a temperature of -18°C . to -20°C . Here, it is noted that a temperature and pressure of the refrigerant passing through the freezing compartment expansion valve **15** may vary depending on the freezing compartment temperature condition.

[0089] When a rear surface of the thermoelectric element **40** is in contact with a front surface of the heat sink **24**, and power is applied to the thermoelectric element **40**, the rear surface of the thermoelectric element **40** becomes a heat generation surface.

[0090] When the cold sink **22** is in contact with a front surface of the thermoelectric element, and power is applied to the thermoelectric element **40**, the front surface of the thermoelectric element **40** becomes a heat absorption surface.

[0091] The cold sink **22** may include a heat conduction plate made of an aluminum material and a plurality of heat exchange fins extending from a front surface of the heat conduction plate. Here, the plurality of heat exchange fins extend vertically and are disposed to be spaced apart from each

other in a horizontal direction.

[0092] Here, when a housing surrounding or accommodating at least a portion of a heat conductor constituted by the heat conduction plate and the heat exchange fin is provided, the cold sink **22** has to be interpreted as a heat transfer member including the housing as well as the heat conductor. This is equally applied to the heat sink **22**, and the heat sink **22** has to be interpreted not only as the heat conductor constituted by the heat conduction plate and the heat exchange fin, but also as the heat transfer member including the housing when a housing is provided.

[0093] The deep freezing compartment fan **25** is disposed in front of the cold sink **22** to forcibly circulate air inside the deep freezing compartment **202**.

[0094] Hereinafter, efficiency and cooling capacity of the thermoelectric element will be described.

[0095] The efficiency of the thermoelectric module **20** may be defined as a coefficient of performance (COP), and an efficiency equation is as follows.

[00001] $COP = \frac{Q_c}{P_e}$ [0096] Q_c : Cooling Capacity (ability to absorb heat) [0097] P_e : Input Power (power supplied to thermoelectric element))

[00002] $P_e = V \times i$

[0098] In addition, the cooling capacity of the thermoelectric module **20** may be defined as follows.

[00003] $Q_c = T_c i - \frac{1}{2} \frac{L}{A} i^2 - \frac{kA}{L} (T_h - T_c)$

<Semiconductor Material Property Coefficient>

[0099] α : Seebeck Coefficient [V/K] [0100] ρ : Specific Resistance [Ωm −1] [0101] k : Thermal Conductivity [W/mk]

<Semiconductor Structure Characteristics>

[0102] L : Thickness of thermoelectric element: Distance between heat absorption surface and heat generation surface [0103] A : Area of thermoelectric element

<System Use Condition>

[0104] i : Current [0105] V : Voltage [0106] $T_{\text{sub.h}}$: Temperature of heat generation surface of thermoelectric element [0107] $T_{\text{sub.c}}$: Temperature of heat absorption surface of thermoelectric element

[0108] In the above cooling capacity equation, a first item at the right may be defined as a Peltier Effect and may be defined as an amount of heat transferred between both ends of the heat absorption surface and the heat generation surface by a voltage difference. The Peltier effect increases in proportional to supply current as a function of current.

[0109] In the formula $V=iR$, since a semiconductor constituting the thermoelectric module acts as resistance, and the resistance may be regarded as a constant, it may be said that a voltage and current have a proportional relationship. That is, when the voltage applied to the thermoelectric module **40** increases, the current also increases. Accordingly, the Peltier effect may be seen as a current function or as a voltage function.

[0110] The cooling capacity may also be seen as a current function or a voltage function. The Peltier effect acts as a positive effect of increasing in cooling capacity. That is, as the supply voltage increases, the Peltier effect increases to increase in cooling capacity.

[0111] The right second item in the cooling capacity equation is defined as a Joule Effect.

[0112] The Joule effect means an effect in which heat is generated when current is applied to a resistor. In other words, since heat is generated when power is supplied to the thermoelectric module, this acts as a negative effect of reducing the cooling capacity. Therefore, when the voltage supplied to the thermoelectric module increases, the Joule effect increases, resulting in lowering of the cooling capacity of the thermoelectric module.

[0113] The right third item in the cooling capacity equation is defined as a Fourier effect.

[0114] The Fourier effect means an effect in which heat is transferred by heat conduction when a temperature difference occurs on both surfaces of the thermoelectric module.

[0115] In detail, the thermoelectric module includes a heat absorption surface and a heat generation

surface, each of which is provided as a ceramic substrate, and a semiconductor element disposed between the heat absorption surface and the heat generation surface. When a voltage is applied to the thermoelectric module, a temperature difference is generated between the heat absorption surface and the heat generation surface. The heat absorbed through the heat absorption surface passes through the semiconductor and is transferred to the heat generation surface. However, when the temperature difference between the heat absorption surface and the heat absorption surface occurs, a phenomenon in which heat flows backward from the heat generation surface to the heat absorption surface by heat conduction occurs, which is referred to as the Fourier effect.

[0116] Like the Joule effect, the Fourier effect acts as a negative effect of lowering the cooling capacity. In other words, when the supply current increases, the temperature difference ($T_h - T_c$) between the heat generation surface and the heat absorption surface of the thermoelectric module, i.e., a value ΔT , increases, resulting in lowering of the cooling capacity.

[0117] FIG. 4 is a graph illustrating a relationship of cooling capacity with respect to the input voltage and the Fourier effect.

[0118] Referring to FIG. 4, the Fourier effect may be defined as a function of the temperature difference between the heat absorption surface and the heat generation surface, that is, a value ΔT .

[0119] In detail, when specifications of the thermoelectric module are determined, values k , A , and L in the item of the Fourier effect in the above cooling capacity equation become constant values, and thus, the Fourier effect may be seen as a function with the value ΔT as a variable.

[0120] Therefore, as the value ΔT increases, the value of the Fourier effect increases, but the Fourier effect acts as a negative effect on the cooling capacity, and thus the cooling capacity decreases.

[0121] As shown in the graph of FIG. 4, it is seen that the greater the value ΔT under the constant voltage condition, the less the cooling capacity.

[0122] In addition, when the value ΔT is fixed, for example, when ΔT is 30° C., a change in cooling capacity according to a change of the voltage is observed. As the voltage value increases, the cooling capacity increases and has a maximum value at a certain point and then decreases again.

[0123] Here, since the voltage and current have a proportional relationship, it should be noted that it is no matter to view the current described in the cooling capacity equation as the voltage and be interpreted in the same manner.

[0124] In detail, the cooling capacity increases as the supply voltage (or current) increases, which may be explained by the above cooling capacity equation. First, since the value ΔT is fixed, the value ΔT becomes a constant. Since the value ΔT for each specification of the thermoelectric element is determined, an appropriate specification of the thermoelectric module may be set according to the required value ΔT .

[0125] Since the value ΔT is fixed, the Fourier effect may be seen as a constant, and the cooling capacity may be simplified into a function of the Peltier effect, which is seen as a first-order function of the voltage (or current), and the Joule effect, which is seen as a second-order function of the voltage (or current).

[0126] As the voltage value gradually increases, an amount of increase in Peltier effect, which is the first-order function of the voltage, is larger than that of increase in Joule effect, which is the second-order function, of voltage, and consequently, the cooling capacity increases. In other words, until the cooling capacity is maximized, the function of the Joule effect is close to a constant, so that the cooling capacity approaches the first-order function of the voltage.

[0127] As the voltage further increases, it is seen that a reversal phenomenon, in which a self-heat generation amount due to the Joule effect is greater than a transfer heat amount due to the Peltier effect, occurs, and as a result, the cooling capacity decreases again. This may be more clearly understood from the functional relationship between the Peltier effect, which is the first-order function of the voltage (or current), and the Joule effect, which is the second-order function of the

voltage (or current). That is, when the cooling capacity decreases, the cooling capacity is close to the second-order function of the voltage.

[0128] In the graph of FIG. 4, it is confirmed that the cooling capacity is maximum when the supply voltage is in a range of 30 V to about 40 V, more specifically, about 35 V. Therefore, if only the cooling capacity is considered, it is said that it is preferable to generate a voltage difference within a range of 30 V to 40V in the thermoelectric module.

[0129] FIG. 5 is a graph illustrating a relationship of efficiency with respect to the input voltage and the Fourier effect.

[0130] Referring to FIG. 5, it is seen that the higher the value ΔT , the lower the efficiency at the same voltage. This will be noted as a natural result because the efficiency is proportional to the cooling capacity.

[0131] In addition, when the value ΔT is fixed, for example, when the value ΔT is limited to 30° C. and the change in efficiency according to the change in voltage is observed, the efficiency increases as the supply voltage increases, and the efficiency decreases after a certain time point elapses. This is said to be similar to the graph of the cooling capacity according to the change of the voltage.

[0132] Here, the efficiency (COP) is a function of input power as well as cooling capacity, and the input P_e becomes a function of V^2 when the resistance of the thermoelectric module **40** is considered as the constant. If the cooling capacity is divided by V^2 , the efficiency may be expressed as Peltier effect-Peltier effect/ V^2 . Therefore, it is seen that the graph of the efficiency has a shape as illustrated in FIG. 5.

[0133] It is seen from the graph of FIG. 5, in which a point at which the efficiency is maximum appears in a region in which the voltage difference (or supply voltage) applied to the thermoelectric module is less than about 20 V. Therefore, when the required value ΔT is determined, it is good to apply an appropriate voltage according to the value to maximize the efficiency. That is, when a temperature of the heat sink and a set temperature of the deep freezing compartment **202** are determined, the value ΔT is determined, and accordingly, an optimal difference of the voltage applied to the thermoelectric module may be determined.

[0134] FIG. 6 is a graph illustrating a relationship of the cooling capacity and the efficiency according to a voltage.

[0135] Referring to FIG. 6, as described above, as the voltage difference increases, both the cooling capacity and efficiency increase and then decrease.

[0136] In detail, it is seen that the voltage value at which the cooling capacity is maximized and the voltage value at which the efficiency is maximized are different from each other. This is seen that the voltage is the first-order function, and the efficiency is the second-order function until the cooling capacity is maximized.

[0137] As illustrated in FIG. 6, as an example, in the case of the thermoelectric module having ΔT of 30° C., it is confirmed that the thermoelectric module has the highest efficiency within a range of approximately 12 V to 17 V of the voltage applied to the thermoelectric module. Within the above voltage range, the cooling capacity continues to increase. Therefore, it is seen that a voltage difference of at least 12 V is required in consideration of the cooling capacity, and the efficiency is maximum when the voltage difference is 14 V.

[0138] FIG. 7 is a view illustrating a reference temperature line for controlling the refrigerator according to a change in load inside the refrigerator.

[0139] Hereinafter, a set temperature of each storage compartment will be described by being defined as a notch temperature. The reference temperature line may be expressed as a critical temperature line.

[0140] A lower reference temperature line in the graph is a reference temperature line by which a satisfactory temperature region and a unsatisfactory temperature region are divided. Thus, a region A below the lower reference temperature line may be defined as a satisfactory section or a satisfactory region, and a region B above the lower reference temperature line may be defined as a

dissatisfied section or a dissatisfied region.

[0141] In addition, an upper reference temperature line is a reference temperature line by which an unsatisfactory temperature region and an upper limit temperature region are divided. Thus, a region C above the upper reference temperature line may be defined as an upper limit region or an upper limit section and may be seen as a special operation region.

[0142] When defining the satisfactory/unsatisfactory/upper limit temperature regions for controlling the refrigerator, the lower reference temperature line may be defined as either a case of being included in the satisfactory temperature region or a case of being included in the unsatisfactory temperature region. In addition, the upper reference temperature line may be defined as one of a case of being included in the unsatisfactory temperature region and a case of being included in the upper limit temperature region.

[0143] When the internal temperature of the refrigerator is within the satisfactory region A, the compressor is not driven, and when the internal temperature of the refrigerator is in the unsatisfactory region B, the compressor is driven so that the internal temperature of the refrigerator is within the satisfactory region.

[0144] In addition, when the internal temperature of the refrigerator is in the upper limit region C, it is considered that food having a high temperature is put into the refrigerator, or the door of the storage compartment is opened to rapidly increase in load within the refrigerator. Thus, a special operation algorithm including a load correspondence operation is performed.

[0145] (a) of FIG. 7 is a view illustrating a reference temperature line for controlling the refrigerator according to a change in temperature of the refrigerating compartment.

[0146] A notch temperature N1 of the refrigerating compartment is set to a temperature above zero. Also, FIG. 2 is a perspective view of the refrigerator door according to an embodiment In order to allow the temperature of the refrigerating compartment to be maintained to the notch temperature N1, when the temperature of the refrigerating compartment rises to a first satisfactory critical temperature N11 higher than the notch temperature N1 by a first temperature difference d1, the compressor is controlled to be driven, and after the compressor is driven, the compressor is controlled to be stopped when the temperature is lowered to a second satisfactory critical temperature N12 lower than the notch temperature N1 by the first temperature difference d1.

[0147] The first temperature difference d1 is a temperature value that increases or decreases from the notch temperature N1 of the refrigerating compartment, and the temperature of the refrigerating compartment may be defined as a control differential or a control differential temperature, which defines a temperature section in which the temperature of the refrigerating compartment is considered as being maintained to the notch temperature N1, i.e., approximately 1.5° C.

[0148] In addition, when it is determined that the refrigerating compartment temperature rises from the notch temperature N1 to a first unsatisfactory critical temperature N13 which is higher by the second temperature difference d2, the special operation algorithm is controlled to be executed. The second temperature difference d2 may be 4.5° C. The first unsatisfactory critical temperature may be defined as an upper limit input temperature.

[0149] After the special driving algorithm is executed, if the internal temperature of the refrigerator is lowered to a second unsatisfactory temperature N14 lower than the first unsatisfactory critical temperature by a third temperature difference d3, the operation of the special driving algorithm is terminated. The second unsatisfactory temperature N14 may be lower than the first unsatisfactory temperature N13, and the third temperature difference d3 may be 3.0° C. The second unsatisfactory critical temperature N14 may be defined as an upper limit release temperature.

[0150] After the special operation algorithm is completed, the cooling capacity of the compressor is adjusted so that the internal temperature of the refrigerator reaches the second satisfactory critical temperature N12, and then the operation of the compressor is stopped.

[0151] (b) of FIG. 7 is a view illustrating a reference temperature line for controlling the refrigerator according to a change in temperature of the freezing compartment.

[0152] A reference temperature line for controlling the temperature of the freezing compartment have the same temperature as the reference temperature line for controlling the temperature of the refrigerating compartment, but the notch temperature N2 and temperature variations k1, k2, and k3 increasing or decreasing from the notch temperature N2 are only different from the notch temperature N1 and temperature variations d1, d2, and d3.

[0153] The freezing compartment notch temperature N2 may be -18°C . as described above, but is not limited thereto. The control differential temperature k1 defining a temperature section in which the freezing compartment temperature is considered to be maintained to the notch temperature N2 that is the set temperature may be 2°C .

[0154] Thus, when the freezing compartment temperature increases to the first satisfactory critical temperature N21, which increases by the first temperature difference k1 from the notch temperature N2, the compressor is driven, and when the freezing compartment temperature is the unsatisfactory critical temperature (upper limit input temperature) N23, which increases by the second temperature difference k2 than the notch temperature N2, the special operation algorithm is performed.

[0155] In addition, when the freezing compartment temperature is lowered to the second satisfactory critical temperature N22 lower than the notch temperature N2 by the first temperature difference k1 after the compressor is driven, the driving of the compressor is stopped.

[0156] After the special operation algorithm is performed, if the freezing compartment temperature is lowered to the second unsatisfactory critical temperature (upper limit release temperature) N24 lower by the third temperature difference k3 than the first unsatisfactory temperature N23, the special operation algorithm is terminated. The temperature of the freezing compartment is lowered to the second satisfactory critical temperature N22 through the control of the compressor cooling capacity.

[0157] Even in the state that the deep freezing compartment mode is turned off, it is necessary to intermittently control the temperature of the deep freezing compartment with a certain period to prevent the deep freezing compartment temperature from excessively increasing. Thus, the temperature control of the deep freezing compartment in a state in which the deep freezing compartment mode is turned off follows the temperature reference line for controlling the temperature of the freezing compartment disclosed in (b) FIG. 7.

[0158] As described above, the reason why the reference temperature line for controlling the temperature of the freezing compartment is applied in the state in which the deep freezing compartment mode is turned off is because the deep freezing compartment is disposed inside the freezing compartment.

[0159] That is, even when the deep freezing compartment mode is turned off, and the deep freezing compartment is not used, the internal temperature of the deep freezing compartment has to be maintained at least at the same level as the freezing compartment temperature to prevent the load of the freezing compartment from increasing.

[0160] Therefore, in the state that the deep freezing compartment mode is turned off, the deep freezing compartment notch temperature is set equal to the freezing compartment notch temperature N2, and thus the first and second satisfactory critical temperatures and the first and second unsatisfactory critical temperatures are also set equal to the critical temperatures N21, N22, N23, and N24 for controlling the freezing compartment temperature.

[0161] (c) of FIG. 7 is a view illustrating a reference temperature line for controlling the refrigerator according to a change in temperature of the deep freezing compartment in a state in which the deep freezing compartment mode is turned on.

[0162] In the state in which the deep freezing compartment mode is turned on, that is, in the state in which the deep freezing compartment is on, the deep freezing compartment notch temperature N3 is set to a temperature significantly lower than the freezing compartment notch temperature N2, i.e., is in a range of about -45°C . to about -55°C ., preferably -55°C . In this case, it is said that

the deep freezing compartment notch temperature N3 corresponds to a heat absorption surface temperature of the thermoelectric module **40**, and the freezing compartment notch temperature N2 corresponds to a heat generation surface temperature of the thermoelectric module **40**.

[0163] Since the refrigerant passing through the freezing compartment expansion valve **15** passes through the deep freezing evaporator, i.e., the heat sink **24**, the temperature of the heat generation surface of the thermoelectric module **40** that is in contact with the heat sink **24** is maintained to a temperature corresponding to the temperature of the refrigerant passing through at least the freezing compartment expansion valve. Therefore, a temperature difference between the heat absorption surface and the heat generation surface of the thermoelectric module, that is, ΔT is 32°C .

[0164] The control differential temperature m1, that is, the deep freezing compartment control differential temperature that defines a temperature section considered to be maintained to the notch temperature N3, which is the set temperature, is set higher than the freezing compartment control differential temperature k1, for example, 3°C .

[0165] Therefore, it is said that the set temperature maintenance consideration section defined as a section between the first satisfactory critical temperature N31 and the second satisfactory critical temperature N32 of the deep freezing compartment is wider than the set temperature maintenance consideration section of the freezing compartment.

[0166] In addition, when the deep freezing compartment temperature rises to the first unsatisfactory critical temperature N33, which is higher than the notch temperature N3 by the second temperature difference m2, the special operation algorithm is performed, and after the special operation algorithm is performed, when the deep freezing compartment temperature is lowered to the second unsatisfactory critical temperature N34 lower than the first unsatisfactory critical temperature N33 by the third temperature difference m3, the special operation algorithm is terminated. The second temperature difference m2 may be 5°C .

[0167] Here, the second temperature difference m2 of the deep freezing compartment is set higher than the second temperature difference k2 of the freezing compartment. In other words, an interval between the first unsatisfactory critical temperature N33 and the deep freezing compartment notch temperature N3 for controlling the deep freezing compartment temperature is set larger than that between the first unsatisfactory critical temperature N23 and the freezing compartment notch temperature N2 for controlling the freezing compartment temperature.

[0168] This is because the internal space of the deep freezing compartment is narrower than that of the freezing compartment, and the thermal insulation performance of the deep freezing case **201** is excellent, and thus, a small amount of the load input into the deep freezing compartment is discharged to the outside. In addition, since the temperature of the deep freezing compartment is significantly lower than the temperature of the freezing compartment, when a heat load such as food is penetrated into the inside of the deep freezing compartment, reaction sensitivity to the heat load is very high.

[0169] For this reason, when the second temperature difference m2 of the deep freezing compartment is set to be the same as the second temperature difference k2 of the freezing compartment, frequency of performance of the special operation algorithm such as a load correspondence operation may be excessively high. Therefore, in order to reduce power consumption by lowering the frequency of performance of the special operation algorithm, it is preferable to set the second temperature difference m2 of the deep freezing compartment to be larger than the second temperature difference k2 of the freezing compartment.

First Embodiment

[0170] FIG. **8** illustrates a cross-sectional view of a thermoelectric module according to a first embodiment.

[0171] Referring to FIG. **8**, a thermoelectric module **20** according to a first embodiment of the present invention includes a thermoelectric element **40**, a cold sink **22** attached to a heat absorption

surface of the thermoelectric element **40**, a heat sink **24** attached to a heat generation surface of the thermoelectric element **40**, and a sealing cover **26** connecting edges of the cold sink **22** and the heat sink **24** to each other.

[0172] As described above, the heat sink **24** may be an evaporator through which a low-temperature refrigerant flows, but it should be noted that, like the cold sink **22**, a metal member having high thermal conductivity may be used.

[0173] In detail, the thermoelectric element **40** may include a semiconductor element portion including a P-type semiconductor element **41** and an N-type semiconductor element **42**, a heat absorption-side electrode **43** provided at one end of the semiconductor element portion, and a heat generation-side electrode **44** provided at the other end of the semiconductor element portion.

[0174] The heat absorption-side electrode **43** and the heat generation-side electrode **44** may be defined as an electrode portion, and the electrode portion may be a copper (Cu) electrode plated with gold (Au). When compared to an electrode portion of a conventional thermoelectric element made of a copper electrode plated with tin (Sn), the copper electrode portion is plated with gold, and thus, solder wetting has been improved. The solder wetting refers to a degree to which a solder material is evenly distributed on a substrate, and the improved solder wetting means that the solder material is spread evenly and thinly on the substrate.

[0175] Thus, as the solder wetting is improved, soldering stability of the semiconductor element portion may be improved.

[0176] In addition, the thermoelectric element **40** includes a heat absorption-side heat dissipation sheet **45** interposed between the heat absorption-side electrode **43** and the cold sink **22** and a heat generation-side heat dissipation sheet **46** interposed between the heat absorption-side electrode **43** and the heat sink **24**.

[0177] The heat absorption-side heat dissipation sheet **45** and the heat generation-side heat dissipation sheet **46** may be defined as a heat dissipation sheet portion, and the heat dissipation sheet portion has a heat transfer function as well as a function of insulating the cold sink **22** or the heat sink **24** from the electrode portion so that electricity does not flow between the cold sink **22** or the heat sink **24** and the electrode portion. That is, it may be understood that the heat dissipation sheet portion replaces a function of a ceramic substrate provided in the conventional thermoelectric element.

[0178] In addition, both ends of the sealing cover **26** are connected to edges of the cold sink **22** and the heat sink **24** by an adhesive **261** to prevent foreign substances such as moisture from being introduced into the thermoelectric element **40**.

[0179] The thermoelectric element **40**, the heat sink **24**, and the cold sink **22** may be formed as a single body by the sealing cover **26**, and as a result, the thermoelectric module may have a thickness that is reduced by a thickness corresponding to that of each of the heat absorption-side and heat generation-side ceramic substrates.

[0180] In case of a general thermoelectric element, the heat absorption-side ceramic substrate attached to the heat absorption-side electrode portion forms the heat absorption surface of the thermoelectric element, and the heat generation-side ceramic substrate attached to the heat generation-side electrode portion forms the heat generation surface of the thermoelectric element.

[0181] The ceramic substrate allows heat to be transferred between the heat absorption-side and heat generation-side electrode surfaces of the thermoelectric element and between the cold sink and the heat sink and also insulates the heat absorption-side and heat generation-side electrode surfaces of the thermoelectric element from each other and insulates the cold sink and the heat sink from each other so that electricity does not flow therebetween.

[0182] The thickness of the ceramic substrate may increase to withstand damage due to external force. For example, the thickness of the ceramic substrate may be greater than that of the electrode portion of the thermoelectric element. As the thickness of the ceramic substrate increases, heat transfer from the electrode portion of the thermoelectric element to the cold sink or the heat sink

may decrease. Thus, if a heat dissipation sheet having a thinner thickness while performing the heat transfer function and the insulating function is used, efficiency of the thermoelectric element may be improved.

[0183] As an example of the heat dissipation sheet according to the present invention, a metal material having strength greater than that of the ceramic may be considered. When a metal having the same material as that of the cold sink or the heat sink is used, thermal resistance generated between metals having different materials may be reduced if the cold sink or the heat sink is designed with an aluminum material, the aluminum material may also be suitable for the heat dissipation sheet. The aluminum material has excellent thermal conductivity higher about 10 times than that of ceramic. In addition, the aluminum material also has an excellent insulating function compared to iron forming a case of the refrigerator.

[0184] In the case of the general thermoelectric element, a sealing material is attached and sealed between a side surface portion of the heat absorption-side ceramic substrate and a side surface portion of the heat generation-side ceramic substrate to prevent foreign substances from being introduced into the semiconductor portion of the thermoelectric element.

[0185] In such a configuration, another heat conduction path may be formed at a position adjacent to the heat absorption surface and the heat generation surface of the thermoelectric element, and thus, heat transfer may occur through the sealing material to reduce efficiency of the thermoelectric element.

[0186] Therefore, it may be advantageous to dispose the sealing material at a place spaced apart from the heat absorption surface and the heat generation surface of the thermoelectric element as necessary. Since the sealing cover **26** according to the present invention is disposed to be spaced apart from the thermoelectric element and is configured to connect and seal the cold sink **22** and the heat sink **24**, the efficiency of the thermoelectric element may be improved.

[0187] In addition, it is possible to improve work convenience by coupling the sealing cover between the cold sink and the heat sink rather than coupling the sealing material between the heat absorption surface and the heat generation surface of the thermoelectric element.

Second Embodiment

[0188] FIG. **9** is a cross-sectional view of a thermoelectric module according to a second embodiment of the present invention.

[0189] Referring to FIG. **9**, in a thermoelectric module **20a** according to a second embodiment of the present invention, two thermoelectric elements **40a** and **40b** are disposed between a cold sink **22** and a heat sink **24**, and the two thermoelectric elements **40a** and **40b** has a structure partitioned by a heat transfer block **27**.

[0190] Also, in the thermoelectric module **20a** according to the present embodiment, like the thermoelectric module **20** according to the first embodiment, a sealing cover **26** is surrounded around an edge of the cold sink **22** and an edge of the heat sink **24** to prevent foreign substances from being introduced into the thermoelectric module **20a**.

[0191] Also, like the thermoelectric module **20** according to the first embodiment, a heat dissipation sheet portion replaces a ceramic substrate.

[0192] In detail, the two thermoelectric elements constituting the thermoelectric module **20a** according to the second embodiment may include a first thermoelectric element **40a** and a second thermoelectric element **40b**, and the first and second thermoelectric elements **40a** and **40b** have the same structure.

[0193] Since the first thermoelectric element **40a** has a heat absorption surface attached to the cold sink **22** to absorb heat, the first thermoelectric element **40a** may be defined as a heat absorption-side thermoelectric element **40a**.

[0194] The second thermoelectric element **40b** has a heat generation surface attached to the heat sink **24** to release the heat transferred through the heat absorption-side thermoelectric element **40a** to the heat sink **24**, the second thermoelectric element **40b** may be defined as a heat generation-side

thermoelectric element **40b**.

[0195] As described above, the thermoelectric module in which two thermoelectric elements are connected to be heat-transferable may be defined as a cascade type thermoelectric element or thermoelectric module. In the cascade type thermoelectric element, the heat generation surface of the first thermoelectric element is connected to be heat-exchangeable with the heat absorption surface of the second thermoelectric element, the cold sink is connected to be heat-transferable to the heat-absorption surface of the first thermoelectric element, and the heat sink is connected to be heat-transferable to the heat generation surface of the second thermoelectric element.

[0196] The technical meaning of the mounting of the heat transfer block **27** may be described in various aspects as follows.

[0197] The heat transfer block **27** may be disposed between the heat generation surface of the first thermoelectric element **40a** and the heat absorption surface of the second thermoelectric element **40b**.

[0198] In another aspect, one surface of the heat transfer block **27** may be coupled to the heat generation surface of the first thermoelectric element **40a**, and the other surface of the heat transfer block **27** may be coupled to the heat absorption surface of the second thermoelectric element **40b**.

[0199] In order to reduce a backflow of heat between the heat generation surface of the first thermoelectric element **40a** and the heat absorption surface of the second thermoelectric element **40b**, the heat transfer block **27** may be disposed between the heat generation surface of the first thermoelectric element **40a** and the heat absorption surfaces of the second thermoelectric element **40b**.

[0200] The heat transfer block **27** may function as a type of heat transfer path formed between the heat generation surface of the first thermoelectric element **40a** and the heat absorption surface of the second thermoelectric element **40b**.

[0201] If there is no heat transfer path, the heat transferred from the cold sink **22** does not flow toward the heat sink **24**, and the heat may be stagnated between the heat absorption surface of the first thermoelectric element **40a** and the heat absorption surface of the second thermoelectric element **40b**.

[0202] If there is no heat transfer path, the heat transferred from the cold sink **22** may not flow toward the heat sink **24** but flow backward. However, there is a disadvantage in that as a longitudinal cross-sectional area of the heat transfer path, which is capable of being defined by a size of a plane crossing a heat flow direction, i.e., a thickness of the heat transfer block increases, thermal resistance increases. In addition, as a length or width of the heat transfer block increases, the thermal resistance may decrease. Here, the length or width of the heat transfer block may be defined as a length extending in a direction crossing the thickness t of the heat transfer block. For example, the length of the heat transfer block may be defined as a length extending in an x-axis direction, and the width of the heat transfer block may be defined on the same plane as the x-axis to extend in a y-axis direction intersecting the x-axis, and the thickness t of the heat transfer block may be defined as a length extending in a z-axis direction intersecting the x-axis and the y-axis.

[0203] The two thermoelectric elements **40a** and **40b** may include semiconductor element portions including P-type semiconductors **41a** and **41b** and N-type semiconductors **42a** and **42b** and electrode portions including heat absorption-side electrodes **43a** and **43b** and heat generation-side electrodes **44a** and **44b**, respectively.

[0204] A thickness of the heat transfer block **27** may be greater than that of the electrode portion of each of the thermoelectric elements **40a** and **40b**. The thickness of the heat transfer block **27** may be less than that of the semiconductor element portion of the thermoelectric device. The thickness of the heat transfer block may be less than each of that of the semiconductor element portion of the first thermoelectric element **40a** and that of the semiconductor element portion of the second thermoelectric element **40b**.

[0205] The length of the heat transfer block **27** may be greater than that of the electrode portion of

each of the thermoelectric elements **40a** and **40b**. The length of the heat transfer block **27** may be greater than that of the semiconductor element portion of the thermoelectric element.

[0206] The heat absorption-side thermoelectric element **40a** may include a pair of heat absorption-side heat dissipation sheets **45a**, and the heat absorption-side thermoelectric element **40b** may include a pair of heat absorption-side heat dissipation sheets **45b**.

[0207] The pair of heat absorption-side heat dissipation sheets **45a** may include a first heat dissipation sheet interposed between the heat absorption-side electrode **43a** and the cold sink **22** and a second heat dissipation sheet interposed therebetween the heat generation-side electrode **44a** and the heat transfer block **27**.

[0208] The pair of heat dissipation sheets **45b** includes a third heat dissipation sheet interposed between the heat transfer block **27** and the heat absorption-side electrode **43b** and a fourth heat dissipation sheet interposed between the heat dissipation electrode **44b** and the heat sink **24**.

[0209] The heat dissipation sheet portion may be understood as an insulating heat transfer sheet that prevents electricity from flowing to the cold sink **22** and the heat sink **24** and allows only the heat transfer.

[0210] The heat transfer block **27** includes a non-electric conductive metal block having a large heat transfer coefficient and may include, for example, an aluminum block having the form of a plate.

[0211] In addition, the heat dissipation sheets **45a** and **45b** may respectively adhere to the cold sink **22**, the heat sink **24**, and the heat transfer block **27** by thermal grease.

[0212] According to the present embodiment, at least two thermoelectric elements may be disposed between the cold sink **22** and the heat sink **24** to divide a temperature difference ΔT between the heat absorption surface of the thermoelectric module **20a**, which is heat-exchanged with the cold sink **22**, and the heat generation surface of the thermoelectric module **20a**, which is heat-exchanged with the heat sink **24**, into a first temperature difference $\Delta T1$ and a second temperature difference $\Delta T2$.

[0213] In detail, in order to achieve the required temperature difference ΔT between the heat absorption surface and the heat generation surface by using one thermoelectric element, power consumption for driving the thermoelectric element increases, and cooling capacity Q_c of the thermoelectric element decreases, and as a result, this causes a problem that the efficiency of the thermoelectric element is deteriorated.

[0214] In order to minimize this problem, two thermoelectric elements **40a** and **40b**, each of which has a small specification in temperature difference ΔT between the heat absorption surface and the heat generation surface may be disposed between the heat sink and the cold sink to cool the storage compartment (e.g., the deep freezing compartment) cooled by the cold sink **22** to a desired temperature and also reduce the power consumption.

[0215] For example, in order to cool the deep freezing compartment to a temperature of -50°C ., when a heat sink **24** through which a refrigerant having a temperature of about -20°C . flows is employed, and a thermoelectric module constituted by a single thermoelectric element is employed, a thermoelectric element having a specification in which the temperature difference ΔT between the heat absorption surface and heat generation surface is 30°C . is required.

[0216] In order to cool the deep freezing compartment to a set temperature using the thermoelectric module, a voltage of about 18V has to be applied to the thermoelectric element. In addition, since the temperature difference is large, the cooling capacity and efficiency are inevitably reduced due to the characteristics of the thermoelectric element.

[0217] However, when using a cascade type thermoelectric element in which two thermoelectric elements having a specification of less than 30°C . are continuously disposed, a temperature difference between the heat sink and the cold sink is maintained at 30°C . to improve the cooling capacity and efficiency of the module.

[0218] A magnitude of a voltage applied to each of the two thermoelectric elements may not be

significantly different from that of a voltage applied to the thermoelectric module including the single thermoelectric element. However, when comparing a time taken to lower the temperature of the heat absorption surface of the thermoelectric module to the set temperature and lower the deep freezing compartment temperature to the set temperature, a time taken when the cascade type thermoelectric module is applied is shorter than that taken when the single thermoelectric element is applied. Thus, it is seen that the cooling capacity of the cascade type thermoelectric module is improved, and the power consumption is reduced, and thus, the efficiency of the thermoelectric module is improved.

[0219] The performance of the cascade type thermoelectric module varies according to the magnitude of the voltage applied to each of the two thermoelectric elements.

[0220] In detail, the thermoelectric element having the same specification, in which the temperature difference $\Delta T1$ between the heat absorption surface and the heat generation surface of the first thermoelectric element **40a** and the temperature difference $\Delta T2$ between the heat absorption surface and the heat generation surface of the second thermoelectric element **40b** have the same value, may be provided, or the thermoelectric element having a different specification, in which the temperature differences are different.

[0221] When an amount of current supplied to the two thermoelectric elements is the same, and the thermoelectric elements having different temperature differences are applied, the temperature difference $\Delta T2$ of the second thermoelectric element **40b** is equal to or less than that $\Delta T1$ of the first thermoelectric element **40a**.

[0222] In detail, since the smaller the temperature difference ΔT , the greater the cooling capacity of the thermoelectric element, it is advantageous to be disposed the thermoelectric element having the relatively small temperature difference ΔT toward the heat sink **24** when the same current is supplied.

[0223] Referring to the equation with respect to the cooling capacity Q_c of the thermoelectric element described above, as the temperature difference decreases, the cooling capacity of the thermoelectric element may increase.

[0224] In other words, the cooling capacity of the second thermoelectric element **40b** is greater than that of the first thermoelectric element **40a**, and thus an amount $Q_{\text{sub.out}}$ of heat transferred to the heat generation surface of the first thermoelectric element **40a** may be completely absorbed from the heat absorption surface of the second thermoelectric element **40b** and transferred to the heat sink **24**.

[0225] When the heat absorption surface of the second thermoelectric element **40b** does not completely absorb the heat transferred from the heat generation surface of the first thermoelectric element **40a**, the efficiency of the thermoelectric module **20a** may decrease.

[0226] For reference, an amount $Q_{\text{sub.out}}$ of heat released from the heat generation surface of the first thermoelectric element **40a** is defined as a sum of the amount Q_c of heat absorbed through the cold sink **22** and an amount P_e of power supplied to the first thermoelectric element **40a** and is expressed as J (Joule) or Kcal.

[0227] On the other hand, when the temperature difference specifications of the first thermoelectric element **40a** and the second thermoelectric element **40b** are set to be the same

[00004]($T1 = T2 = \frac{T}{2}$),

amounts of current (or voltage difference) applied to the first thermoelectric element **40a** and the second thermoelectric element **40b** may be differently controlled to maximally improve the efficiency of the thermoelectric module **20a**.

[0228] First, a case in which a voltage having the same magnitude is applied to the first thermoelectric element **40a** and the second thermoelectric element **40b** will be described. The voltage having the same magnitude may be any one of the same high voltage, the same medium voltage, and the same low voltage.

[0229] When a predetermined voltage V_a is supplied to the first thermoelectric element **40a**, an

amount $Q_{\text{sup.out1}}$ of heat released to the heat generation surface of the first thermoelectric element **40a** is as follows

$$[00005] Q_{\text{out1}} = Q_{c1} + P_{e1}$$

[0230] When a predetermined voltage V_a is supplied to the first thermoelectric element **40a** and the second thermoelectric element **40b**, an amount $Q_{\text{sub.out2}}$ of heat that has to be released through the heat sink **24** attached to the heat generation surface of the second thermoelectric element **40b** is as follows.

$$[00006] Q_{\text{out2}} = Q_{\text{out1}} + P_{e2}, [0231] P_{\text{sub.e2}}: \text{Amount of power supplied to the second thermoelectric element } \mathbf{40b}$$

[0232] Here, when the heat sink **24** is an evaporator through which the refrigerant passing through the freezing compartment expansion valve flows, a surface temperature of the heat sink **24** is maintained at a temperature in a range of approximately -25°C. to -30°C.

[0233] In this case, in order that the heat sink **24** quickly absorbs the heat amount $Q_{\text{sub.out2}}$, the temperature of the refrigerant passing through the heat sink **24** has to be lowered, or the heat sink **24** has to be designed to be very large. That is, when a voltage having the same magnitude is applied to the two thermoelectric elements, heat absorbed from the cold sink **22** is not rapidly released through the heat sink **24**.

[0234] In another aspect, when the heat sink **24** is designed to be very large, there may be a disadvantage in that a volume of the storage compartment of the refrigerator is relatively reduced.

[0235] In addition, in order to further lower the temperature of the refrigerant passing through the heat sink **24**, it may be difficult to develop a new refrigerant having efficiency greater than the current refrigerant.

[0236] In addition, in order to further lower the temperature of the refrigerant passing through the heat sink **24**, there may be a disadvantage in that the components of the refrigerant circulation system such as the compressor and the condenser of the refrigerator have to be designed to be larger. Due to this, the volume of the storage compartment of the refrigerator may be further reduced.

[0237] Thus, while the power consumption increases, the cooling capacity of the thermoelectric module **20a** decreases to deteriorate the efficiency of the thermoelectric module **20a**.

[0238] In this problem, when the voltage applied to the first thermoelectric element **40a** is greater than the voltage applied to the second thermoelectric element **40b**, since the heat released from the heat generation surface of the second thermoelectric module **40b** through the heat generation surface of the first thermoelectric element **40a** may not be completely absorbed to further deteriorate the efficiency of the thermoelectric module **20a**.

[0239] On the other hand, when the voltage applied to the second thermoelectric element **40b** is greater than the voltage applied to the first thermoelectric element **40a**, the heat absorbed through the heat absorption surface of the first thermoelectric element **40a** may be completely absorbed by the second thermoelectric element and then be released to the outside of the thermoelectric module **20a** through the heat sink **24**.

[0240] Particularly, when the high voltage is applied to the second thermoelectric element **40b**, any one of the medium voltage and the low voltage may be applied to the first thermoelectric element **40a**, and when the medium voltage is applied to the second thermoelectric element **40b**, the low voltage may be applied to the first thermoelectric element **40a** in terms of improving the efficiency of the thermoelectric module **20a**.

[0241] It should be noted that the power control or voltage control of the cascade type thermoelectric module including the two thermoelectric elements described above is equally applied to the cascade type thermoelectric module according to the third and fourth embodiments described below.

[0242] The thermoelectric module according to the present invention may include a cold sink **22**, a first thermoelectric element **40a** having a heat absorption surface connected to the cold sink, a

second thermoelectric element having a heat absorption surface connected to a heat generation surface of the first thermoelectric element **40a**, and a heat sink **24** connected to a heat generation surface of the second thermoelectric element **40b** to release heat transferred from the cold sink **22** to the outside.

[0243] The first thermoelectric element **40a** may include a semiconductor element portion including a P-type semiconductor and an N-type semiconductor, a heat absorption-side electrode portion **44a** formed at one end of the semiconductor element portion and connected to the cold sink, and a heat generation-side electrode portion **43a** formed at the other end of the semiconductor element portion and connected to the heat absorption surface of the second thermoelectric element.

[0244] The second thermoelectric element **40b** may include a semiconductor element portion including a P-type semiconductor and an N-type semiconductor, a heat absorption-side electrode portion **44b** formed at one end of the semiconductor element portion and connected to the heat generation surface of the first thermoelectric element **40a**, and a heat generation-side electrode portion **43b** formed at the other end of the semiconductor element portion and connected to the heat sink.

[0245] The thermoelectric module may further include a sealing cover connecting an edge of the cold sink to an edge of the heat sink.

[0246] Such a configuration may improve work convenience.

[0247] The sealing cover may be configured to seal the edge of the cold sink and the edge of the heat sink. This configuration may prevent foreign substances from being introduced into the thermoelectric element.

[0248] The sealing cover may be disposed to be spaced apart from the first and second thermoelectric elements **40a** and **40b**. This configuration may improve the heat transfer efficiency of the thermoelectric module.

[0249] The thermoelectric module may further include a first heat dissipation sheet **45a** connected between the cold sink **22** and the heat absorption-side electrode portion of the first thermoelectric element **40a** and a second heat dissipation sheet **45b** connected between the heat sink **24** and the heat generation-side electrode portion of the second thermoelectric element **40b**.

[0250] The first and second heat dissipation sheets **45a45b** may perform a heat transfer function and an insulating function.

[0251] The thermoelectric module may further include a heat transfer block **27** interposed between the first thermoelectric element **40a** and the second thermoelectric element **40b**.

[0252] The heat absorption-side electrode portion of the first thermoelectric element **40a** may be attached to the cold sink, and the heat generation-side electrode portion of the second thermoelectric element **40b** may be attached to the heat sink **24**.

[0253] The heat generation-side electrode portion of the first thermoelectric element **40a** may be attached to one surface of the heat transfer block **27**, and the heat absorption-side electrode portion of the second thermoelectric element **40b** may be attached to the other surface of the heat transfer block **27**.

[0254] The heat transfer block **27** may reduce heat transferred from the cold sink **22** from flowing backward instead of flowing toward the heat sink **24**.

[0255] The controller may control a voltage so that either the high voltage or the low voltage lower than the high voltage is applied to the first thermoelectric element **40a** and the second thermoelectric element **40b**.

[0256] The controller may control a voltage applied to the second thermoelectric element **40b** to be greater than a voltage applied to the first thermoelectric element **40a**. This configuration may reduce the increase in size of the heat sink **24** as well as reduce the increase in means for cooling the heat sink **24**.

[0257] Thus, it is possible to reduce an increase in volume of the evaporation compartment of the refrigerator, and as a result, it is possible to increase in volume of the storage compartment of the

refrigerator.

[0258] The temperature difference ΔT_1 between the heat absorption surface and the heat generation surface of the first thermoelectric element may be equal to or greater than the temperature difference ΔT_2 between the heat absorption surface and the heat generation surface of the second thermoelectric element. This is because it is advantageous to design the thermoelectric element adjacent to the heat sink to have large cooling capacity.

Third Embodiment

[0259] FIG. **10** is a cross-sectional view of a thermoelectric module according to a third embodiment of the present invention.

[0260] Referring to FIG. **10**, in a thermoelectric module **20b** according to a third embodiment of the present invention, as in the second embodiment, the ideas in which two thermoelectric modules **51** and **52** is disposed between a cold sink **22** and a heat sink **24** is the same. However, there is a slight difference in configuration of the two thermoelectric modules **51** and **52**.

[0261] In detail, the thermoelectric module **20b** according to the third embodiment may include a cold sink **22**, a first thermoelectric element **51** having a heat absorption surface attached to the cold sink **22**, a second thermoelectric element **52** having a heat absorption surface that is thermally conductively connected to a heat generation surface of the first thermoelectric element **51**, a heat sink **24** attached to a heat generation surface of the second thermoelectric element **52**, and an intermediate substrate **53** interposed between the first thermoelectric element **51** and a bonding portion of the second thermoelectric element **52**.

[0262] In more detail, the first thermoelectric element **51** may include a semiconductor element portion, an electrode portion, a heat absorption-side substrate **515**, and a sealing member **516**.

[0263] The semiconductor element portion includes a P-type semiconductor element **511** and an N-type semiconductor element **512**.

[0264] The electrode unit includes a heat absorption-side electrode **513** provided at one end of the semiconductor element portion and a heat generation-side electrode **514** provided at the other end of the semiconductor element portion.

[0265] One surface of the heat absorption-side substrate **515** is attached to the cold sink **22** by thermal grease **517**, and the heat absorption-side electrode **513** is formed on the other surface of the heat absorption-side substrate **515**.

[0266] In addition, the sealing member **516** is surrounded round edges of the heat absorption-side substrate **515** and the intermediate substrate **53**.

[0267] The second thermoelectric element **52** includes a semiconductor element portion, an electrode portion, a heat generation-side substrate **525**, and a sealing member **526**.

[0268] The semiconductor element portion includes a P-type semiconductor element **521** and an N-type semiconductor element **522**.

[0269] The electrode unit includes a heat absorption-side electrode **523** provided at one end of the semiconductor element portion and a heat generation-side electrode **524** provided at the other end of the semiconductor element portion.

[0270] One surface of the heat generation-side substrate **525** is attached to the heat sink **24** by the thermal grease **54**, and the heat generation-side electrode **523** is formed on the other surface of the heat generation-side substrate **525**.

[0271] In addition, the sealing member **526** is surrounded around the edges of the heat generation-side substrate **525** and the intermediate substrate **53**.

[0272] A heat generation-side electrode **514** of the first thermoelectric element **51** is formed on one surface of the intermediate substrate **53**, and a heat absorption-side electrode **523** of the second thermoelectric element **52** is formed on the other surface of the intermediate substrate **53**.

[0273] The heat absorption-side substrate **515**, the heat generation-side substrate **525**, and the intermediate substrate **53** may be made of a ceramic material, like the conventional thermoelectric device. In addition, the sealing members **516** and **526** may also be sealing members of the

conventional thermoelectric element for preventing the foreign substances from being introduced into the thermoelectric element by connecting the edges of the ceramic substrate.

[0274] According to this embodiment, although the conventional structure in which the two thermoelectric elements are connected to each other, it is characterized in that the single intermediate substrate **53** is coupled to the bonding portion of the two thermoelectric elements.

Fourth Embodiment

[0275] FIG. **11** is a cross-sectional view of a thermoelectric module according to a fourth embodiment of the present invention.

[0276] Referring to FIG. **11**, a thermoelectric module **20c** according to a fourth embodiment of the present invention has the same structure as the thermoelectric module **20b** according to the third embodiment, except that two intermediate substrates are applied.

[0277] In detail, in the thermoelectric module **20c** according to the fourth embodiment, a first thermoelectric element **53** and a second thermoelectric element **53** are attached to each other by thermal grease **54** and then attached to a cold sink **22** and a heat sink **24**.

[0278] The first thermoelectric element **53** and the second thermoelectric element **54** have the same structure as the conventional thermoelectric element structure.

[0279] The first thermoelectric element **53** includes a heat absorption-side substrate **535**, a heat generation-side substrate **536**, a semiconductor element portion, an electrode portion, and a sealing member **537**. The semiconductor element portion includes a P-type semiconductor element **531** and an N-type semiconductor element **532**.

[0280] The electrode portion includes a heat absorption-side electrode **533** formed between one end of the semiconductor element portion and the heat absorption-side substrate **535** and a heat generation-side electrode **534** formed between the other end of the semiconductor element portion and the heat generation-side substrate **536**. Since the second thermoelectric element **53** has the same structure as the first thermoelectric element **53**, a duplicated description thereof will be omitted.

[0281] In addition, the heat generation-side substrate **536** of the first thermoelectric element **53** is attached to the heat absorption-side substrate of the second thermoelectric element **54** by the thermal grease **54**.

[0282] The heat absorption-side substrate **535** of the first thermoelectric element is coupled to the cold sink **22** by the thermal grease **54**.

[0283] The heat generation-side substrate of the second thermoelectric element is coupled to the heat sink **24** by the thermal grease **54**.

[0284] In addition, edges of the heat absorption-side substrate **535** and the heat generation-side substrate **536** constituting the first thermoelectric element **53** are connected to each other by the sealing member **537** so that a space in which the semiconductor element portion is accommodated is separated from the outside. This may be also applied to the second thermoelectric element **54**.

Fifth Embodiment

[0285] FIG. **12** is a cross-sectional view of a thermoelectric module according to a fifth embodiment of the present invention.

[0286] Referring to FIG. **12**, a thermoelectric module **20d** according to a fifth embodiment is substantially the same as the structure of the thermoelectric module **20** according to the first embodiment illustrated in FIG. **8**, except for the following portions.

[0287] First, a heat transfer block **27** is interposed between a heat absorption-side heat dissipation sheet **45** and a cold sink **22** and between a heat generation-side heat dissipation sheet **46** and a heat sink **24**, respectively. Also, heat transfer blocks **27** are respectively fixed to a cold sink **22** and a heat sink **24** by thermal grease **54**.

[0288] Second, both ends of the sealing cover **26** are respectively connected to side surfaces of the two heat transfer blocks **27** to block foreign substances from being introduced into the thermoelectric element.

[0289] Here, a heat insulator **23** is interposed between the cold sink **22** and the heat sink **24** to block heat transfer from the heat sink **24** to the cold sink **22**.

[0290] Since the thermoelectric element **40** according to the present embodiment has the same configuration as that of the thermoelectric element **40** according to the first embodiment, a duplicated description thereof will be omitted.

Sixth Embodiment

[0291] FIG. **13** is a cross-sectional view of a thermoelectric module according to a sixth embodiment of the present invention.

[0292] Referring to FIG. **13**, a thermoelectric module **20e** according to a sixth embodiment of the present invention has the same configuration as that of the thermoelectric module **20a** according to the second embodiment shown in FIG. **9**, except for the following differences.

[0293] First, a heat transfer block **27** is interposed between a cold sink **22** and a first thermoelectric element **40a** and between a heat sink **24** and a second thermoelectric element **40b**, respectively. The heat transfer blocks **27** may be fixed to the cold sink **22** and the heat sink **24** by thermal grease **54**, respectively.

[0294] Thus, the thermoelectric module **20e** according to the sixth embodiment is different from the thermoelectric module **20a** according to the second embodiment in that all three heat transfer blocks **27** are provided.

[0295] Here, the heat transfer block attached to the cold sink may be defined as a first heat transfer block, the heat transfer block attached to the heat sink may be defined as a second heat transfer block, and the heat transfer block interposed between the first thermoelectric element **40a** and the second thermoelectric element **40b** may be defined as a third heat transfer block.

[0296] Second, both ends of a sealing cover **26** are coupled to side surfaces of the first heat transfer block and the second heat transfer block. This is substantially the same as the structure of the sealing cover **26** described in the fifth embodiment.

[0297] In the sixth embodiment, like the fifth embodiment, a heat insulator **23** may be interposed between the cold sink **22** and the heat sink **24** to prevent backflow of heat from the heat sink **24** to the cold sink **22**.

[0298] In summary, the cascade type thermoelectric modules disclosed in the second to fourth embodiments and the sixth embodiment have the following common characteristics.

[0299] First, the thermoelectric module includes a first thermoelectric element and a second thermoelectric element.

[0300] Second, a heat absorption surface of the first thermoelectric element is thermally conductively connected to the cold sink, and a heat generation surface of the second thermoelectric element is thermally conductively connected to the heat sink.

[0301] Third, a heat generation surface of the first thermoelectric element is thermally conductively connected to a heat absorption surface of the second thermoelectric element.

[0302] Fourth, magnitudes of a voltage applied to the first thermoelectric element and a voltage applied to the second thermoelectric element may be different. Specifically, the voltage applied to the second thermoelectric element may be greater than the voltage applied to the first thermoelectric element.

[0303] Fifth, a space between ceramic substrates is shielded or sealed from an external space by a sealing member, or a space between the cold sink and the heat sink is shielded or sealed from the external space by a sealing cover.

[0304] In addition, the thermoelectric modules according to the first to sixth embodiments have the following common characteristics.

[0305] That is, both ends of the semiconductor element portion are soldered to a heat dissipation sheet or a substrate made of a ceramic material by an electrode portion made of copper material, and a plating layer made of a gold (Au) material is formed on the electrode portion to improve solder wetting, thereby improving soldering stability of the semiconductor element portion.

Claims

1. A thermoelectric module comprising: a cold sink including a plate portion; a first thermoelectric element of which a heat absorption surface is coupled to the cold sink; a heat sink including a plate portion; a second thermoelectric element disposed between a heat generation surface of the first thermoelectric element and the heat sink, a heat generation surface of the second thermoelectric element being coupled to the plate portion of the heat sink to release heat transferred from the cold sink to an outside of the second thermoelectric element; a heat transfer block interposed between the heat generation surface of the first thermoelectric element and a heat absorption surface of the second thermoelectric element such that the heat transfer block is configured to function as a type of heat transfer path formed between the heat generation surface of the first thermoelectric element and the heat absorption surface of the second thermoelectric element; a sealing cover directly connecting an edge of the cold sink and an edge of the heat sink to each other, to seal the first and second thermoelectric elements and to prevent a foreign substance from being introduced into the first and second thermoelectric elements, wherein the first thermoelectric element comprises: a semiconductor element portion comprising a P-type semiconductor and an N-type semiconductor; a heat absorption-side electrode portion disposed at one end of the semiconductor element portion; a heat generation-side electrode portion disposed at the other end of the semiconductor element portion; a first heat dissipation sheet disposed between the plate portion of the cold sink and the heat absorption-side electrode portion; and a second heat dissipation sheet disposed between a first surface of the heat transfer block and the heat generation-side electrode portion, wherein each of the first and second heat dissipation sheets includes a metal sheet having strength greater than that of a ceramic, wherein the cold sink is provided with a plurality of heat exchange fins, and wherein the heat sink is provided with at least one of a plurality of heat exchange fins or an evaporator through which a refrigerant flows, wherein the sealing cover is connected edges of the cold sink and the heat sink by an adhesive, wherein the sealing cover surrounds edges of the first and second thermoelectric elements and the heat transfer block.
2. The thermoelectric module according to claim 1, wherein the semiconductor element portion is soldered on the respective first and second heat dissipation sheets.
3. The thermoelectric module according to claim 2 wherein the second thermoelectric element includes: a semiconductor element portion comprising a P-type semiconductor and an N-type semiconductor; a heat absorption-side electrode portion disposed at one end of the semiconductor element portion; a heat generation-side electrode portion disposed at the other end of the semiconductor element portion; a third heat dissipation sheet disposed between a second surface of the heat transfer block and the heat absorption-side electrode portion; and a fourth heat dissipation sheet disposed between the heat generation-side electrode portion and the heat sink, wherein each of the third and fourth heat dissipation sheets of the second thermoelectric element includes a metal seat having strength greater than that of the ceramic.
4. The thermoelectric module according to claim 1, wherein the heat transfer block comprises an aluminum metal block.
5. The thermoelectric module according to claim 1, wherein any one voltage of a high voltage, a medium voltage lower than the high voltage, and a low voltage lower than the medium voltage is applied to the first thermoelectric element and the second thermoelectric element, and the voltage applied to the first thermoelectric element and the voltage applied to the second thermoelectric element are different voltages.
6. The thermoelectric module according to claim 5, wherein the voltage applied to the second thermoelectric element is greater than the voltage applied to the first thermoelectric element.
7. The thermoelectric module according to claim 6, wherein, when the voltage applied to the second thermoelectric element is the high voltage, the voltage applied to the first thermoelectric

element is the medium voltage or the low voltage.

8. The thermoelectric module according to claim 1, wherein a temperature difference between the heat absorption surface and the heat generation surface of the first thermoelectric element is set to be the same as a temperature difference between the heat absorption surface and the heat generation surface of the second thermoelectric element.

9. The thermoelectric module according to claim 1, wherein the sealing cover is connected to an edge of the plate portion of the cold sink such that the plurality of heat exchange fins is exposed outside the sealing cover.

10. The thermoelectric module according to claim 3, wherein a length of the heat transfer block is greater than that of the heat absorption-side electrode portion of each of the first and second thermoelectric elements and that of the heat generation-side electrode portion of each of the first and second thermoelectric elements, and/or wherein a length of the heat transfer block is greater than that of the semiconductor element portion of each of the first and second thermoelectric elements.

11. The thermoelectric module according to claim 3, wherein a thickness of the heat transfer block is greater than that of the heat absorption-side electrode portion of each of the first and second thermoelectric elements and that of the heat generation-side electrode portion of each of the first and second thermoelectric elements, and/or wherein a thickness of the heat transfer block is less than that of the semiconductor element portion of each of the first and second thermoelectric elements.

12. The thermoelectric module according to claim 1, wherein the sealing cover including an inner surface spaced apart from an edge of the heat transfer block.

13. The thermoelectric module according to claim 11, wherein a size of the heat sink is greater than a size of the cold sink.

14. A thermoelectric module comprising: a cold sink including a plate portion, a first thermoelectric element of which a heat absorption surface is coupled to the cold sink; a heat sink including a plate portion; a second thermoelectric element disposed between a heat generation surface of the first thermoelectric element and the heat sink, a heat generation surface of the second thermoelectric element being coupled to the plate portion of the heat sink to release heat transferred from the cold sink to an outside of the second thermoelectric element; a heat transfer block interposed between the heat generation surface of the first thermoelectric element and a heat absorption surface of the second thermoelectric element such that the heat transfer block is configured to function as a type of heat transfer path formed between the heat generation surface of the first thermoelectric element and the heat absorption surface of the second thermoelectric element; a first heater transfer block interposed between the plate portion of the cold sink and the heat absorption surface of the first thermoelectric element; a second heater transfer block interposed between the plate portion of the heat sink and the heat generation surface of the second thermoelectric element; and a sealing cover directly connecting an edge of the plate portion of the cold sink and an edge of the heat sink to each other, to seal the first and second thermoelectric elements, and to prevent a foreign substance from being introduced into the first and second thermoelectric element, wherein the sealing cover is connected edges of the cold sink and the heat sink by an adhesive, wherein the sealing cover surrounds edges of the first and second thermoelectric element, the heat transfer block, the first heater transfer block and the second heater transfer block.

15. The thermoelectric module according to claim 14, wherein the first thermoelectric element comprises: a first heat dissipation sheet disposed between the first heater transfer block and a heat absorption-side electrode portion; and a second heat dissipation sheet disposed between a first surface of the heat transfer block and a heat generation-side electrode portion, wherein the first heater transfer block is interposed between the plate portion of the cold sink and the first heat dissipation sheet.

16. The thermoelectric module according to claim 15, wherein the second thermoelectric element

includes: a third heat dissipation sheet disposed between a second surface of the heat transfer block and a heat absorption-side electrode portion; and a fourth heat dissipation sheet disposed between a heat generation-side electrode portion and the second heater transfer block, wherein the second heater transfer block is interposed between the plate portion of the heat sink and the fourth heat dissipation sheet.

17. The thermoelectric module according to claim 14, wherein the sealing cover including an inner surface spaced apart from edges of the heat transfer block, the first heater transfer block and the second heater transfer block.

18. A thermoelectric module comprising: a cold sink including a plate portion and a plurality of heat exchange fins; a first thermoelectric element of which a heat absorption surface is coupled to the cold sink; a heat sink including a plate portion; a second thermoelectric element disposed between a heat generation surface of the first thermoelectric element and the heat sink, a heat generation surface of the second thermoelectric element being coupled to the plate portion of the heat sink to release heat transferred from the cold sink to an outside of the second thermoelectric element; a heat transfer block interposed between the heat generation surface of the first thermoelectric element and a heat absorption surface of the second thermoelectric element such that the heat transfer block is configured to function as a type of heat transfer path formed between the heat generation surface of the first thermoelectric element and the heat absorption surface of the second thermoelectric element; a sealing cover directly connecting an edge of the plate portion of the cold sink and an edge of the heat sink to each other, to seal the first and second thermoelectric elements and to prevent a foreign substance from being introduced into the first and second thermoelectric elements, and to expose the plurality of heat exchange fins outside the sealing cover, wherein the sealing cover is connected edges of the cold sink and the heat sink by an adhesive, wherein the sealing cover surrounds edges of the first and second thermoelectric elements, and the heat transfer block.

19. The thermoelectric module according to claim 18, wherein the sealing cover including an inner surface spaced apart from an edge of the heat transfer block.

20. The thermoelectric module according to claim 18, wherein a size of of the heat sink is greater than a size of the cold sink.
